

SUMMARY

AI Engineer with hands-on experience building end-to-end generative AI systems - RAG pipelines, LLM-powered APIs, agentic workflows, and NL-to-SQL engines from design through deployment. Proficient in LangChain, LangGraph, Vertex AI (Gemini), Groq, ChromaDB, and pgvector. Skilled in designing scalable APIs, implementing LLM evaluation frameworks, diagnosing hallucination failures, and shipping reliable AI systems with strong software engineering practices.

TECHNICAL SKILLS

Generative AI & LLMs: LangChain (Chains, Agents, LCEL), Agentic Workflows (LangGraph, ReAct), Vertex AI (Gemini), LangSmith, OpenAI API, Groq (Llama 3.3), RAG Pipelines, Prompt Engineering, LLM Evaluation (BLEU, ROUGE), Hallucination Detection & Mitigation

AI/ML Engineering: Scikit-learn, XGBoost, SHAP, SMOTE, NLP (spaCy, NLTK, TF-IDF, NER), Feature Engineering, Cross-Validation, Hyperparameter Tuning, Model Evaluation

Data Engineering: Python, Pandas, ETL Pipeline Design, RapidFuzz (Fuzzy Matching), Data Validation & Quality Assurance, Schema Validation, REST API Integration (U.S. Census Bureau API)

Vector Databases & Embeddings: ChromaDB, PostgreSQL (pgvector), HuggingFace Embeddings (all-MiniLM-L6-v2), Vertex AI Embeddings, Semantic Search, Similarity Retrieval

Software & APIs: Python, FastAPI, RESTful API Design, Pydantic, SQL, pytest, Git/GitHub, CI/CD (GitHub Actions), Docker, Error Handling, Rate Limiting, Debugging

Cloud & MLOps: GCP (Vertex AI, Cloud Run, Cloud Logging, BigQuery), Docker & Compose, Streamlit, Model Serving, Containerization

BI & Visualization: Tableau, Power BI (DAX, PL-300 Certified), Matplotlib, Seaborn

EXPERIENCE

Project Assistant – University of South Florida

Feb 2025 – May 2026

- Designed and deployed a Python-based ETL pipeline standardizing 180+ country name variations (EU, USSR, UAE) into ISO codes using RapidFuzz and custom override dictionaries, reducing mismatch errors by ~90%.
- Automated ingestion, preprocessing, feature engineering, schema validation, and metadata generation through reusable modules for quality assurance (anomaly detection, duplicate identification, exception handling) and semantic search/RAG-ready preparation, cutting manual effort by ~70% and improving reliability across ML and retrieval-based workflows.
- Integrated the U.S. Census Bureau ACS 5-Year API using FIPS-based conditional logic to extract demographic and income data at ZIP, county, and state granularity, producing analytics-ready, LLM-ready datasets for BI and AI applications.
- Implemented Git-based CI/CD workflows to automate testing and validation of data pipelines, ensuring reproducibility and enabling collaboration with faculty and analytics teams.

Data Science Intern - Accenture

Jun 2023 – May 2024

- Built and optimized ML data pipelines for cognitive and socioeconomic datasets (5M+ records), feature engineering, EDA, and data quality resolution directly improving model performance across 5+ analytics teams
- Developed Tableau and Power BI dashboards translating ML model outputs into actionable KPIs for C-level stakeholders; automated reporting reducing manual reporting time by 30%
- Built SQL-based reporting pipelines analyzing operational trends and improving feature consistency across ML workflows; designed and documented data models for 8+ downstream analytics use cases

Research Assistant – Malla Reddy Engineering College for women

Jun 2022 – Jun 2023

- Analyzed 18,617 OSHA accident reports using NLP (spaCy, TF-IDF, NER) trained 4 ML classifiers (SVM, Random Forest, Gradient Boosting, Logistic Regression); best model achieved 96% accuracy and AUC-ROC of 0.9867
- Integrated Gemini LLM to generate real-time safety recommendations and OSHA standard citations for high-risk cases; deployed end-to-end system as Streamlit + Docker application
- Built and deployed multi-disease classification system (heart disease, diabetes, Parkinson's) using XGBoost, Random Forest with cross-validation and hyperparameter tuning - achieving 89%+ accuracy across all three conditions

PROJECTS

RAG-Based Document Q&A System | LangChain, FastAPI, ChromaDB, pgvector, Vertex AI, Groq, Streamlit, Pytest

- Built and deployed production RAG system end-to-end: PDF ingestion, HuggingFace embedding generation (all-MiniLM-L6-v2), ChromaDB vector storage, Groq LLM (Llama 3.3) answer generation - live on Hugging Face Spaces with <500ms latency
- Architected dual-backend design: pgvector (PostgreSQL) for enterprise deployments + ChromaDB for lightweight deployments; FastAPI endpoints with Pydantic validation, rate limiting, and Cloud Logging for production monitoring
- Implemented LLM evaluation framework measuring answer relevance (ROUGE), fact consistency, and retrieval quality; diagnosed and resolved hallucination failures through systematic retrieval tuning and prompt engineering
- Authored end-to-end integration documentation covering embedding config, vector DB tuning, API setup, and LLM evaluation; built pytest unit/integration test suite with performance benchmarks

AI-Powered SQL Analytics Assistant | LangGraph, Groq (Llama 3.3 70B), Python, SQLite, Streamlit, SQL

- Built and deployed an agentic NL-to-SQL pipeline using LangGraph ReAct agents with tool orchestration for SQL generation, validation/execution, and natural language explanation, hosted on Hugging Face Spaces.
- Powered system with Llama 3.3 70B via Groq, integrating conversational memory for multi-turn queries and session-aware context retention without requiring query restatement.
- Engineered schema-aware prompting with live database metadata injection and SQL guardrails (SELECT-only enforcement, JOIN validation, LIMIT constraints), achieving 95%+ query accuracy while preventing unsafe execution.
- Designed a Streamlit-based chat interface with auto chart generation and toggleable learning modes (prompt, schema, SQL, validation), enabling non-technical users to interact with databases without SQL knowledge.

EDUCATION

University of South Florida - Master of Science, Artificial Intelligence and Business Analytics, Tampa, FL

Aug 2024 – May 2026

Malla Reddy Engineering College for Women - Bachelor of Technology, Data Science, Hyderabad, India

Jun 2020 – Apr 2024

CERTIFICATIONS

Data Science Fundamentals (Python/SQL) & Intro to Data Science – IBM | Power BI Data Analyst (PL-300) | Programming Essentials – Cisco | Python (Basic) – HackerRank